GLY 421 SEDIMENTARY BASIN ANALYSIS

Sedimentary basin analysis is the aspect of geology that considers all the controls on the accumulation of a succession of sedimentary rocks to develop a model for the evolution of the sedimentary basin as a whole (Fig. 1)

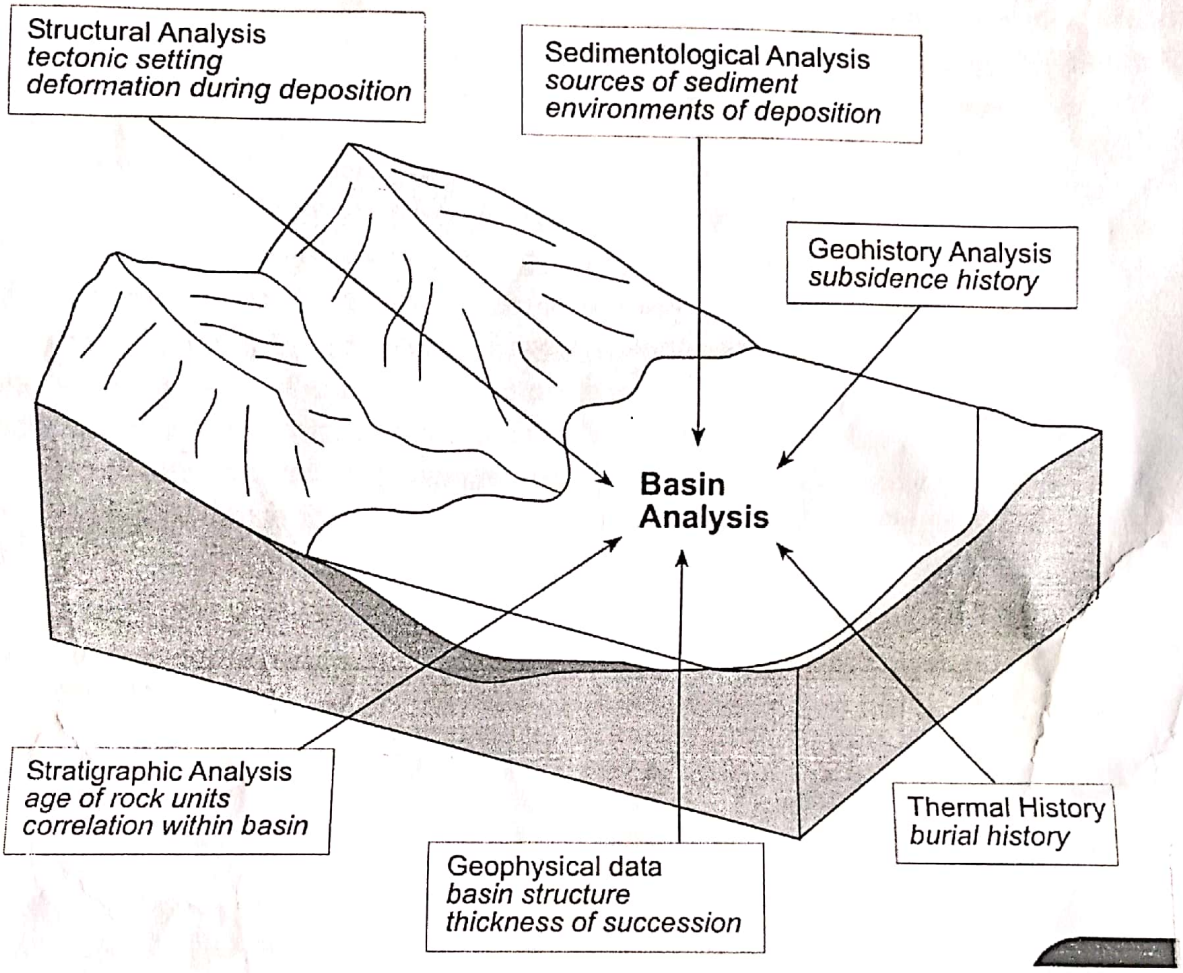


Fig. 1: Basin analysis techniques

Structural analysis

The patterns of deformation within a sedimentary succession provide information about the crustal stresses that existed in the area during and after deposition. Synsedimentary faults and folds are evidence of tectonic activity during deposition: a layer of strata may show structures (such as normal faults) that indicate extension during sedimentation or evidence of compressive forces (reverse faults or folds) acting as the strata were accumulating. These growth structures can be identified by the fact that their occurrence is limited to a particular stratigraphic unit within the succession. In general, extensional settings such as rift basins can be distinguished from basins formed under compressional regimes (such as foreland basins) on the basis of the recognition of these syndepositional structures, although local variations in stress commonly occur. Structural features that affect the whole succession are evidence

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of tectonic forces occurring after deposition in the basin, but nevertheless provide information about the plate tectonics history of the area.

Geophysical data

Information from seismic reflection surveys provides subsurface structural data that can be used in the structural analysis of a succession. Other geophysical data are in the form of information about the magnetic properties of the rocks below the surface and variations in the strength of the gravitational field in the region of the basin. Magnetic surveys over an area can indicate the nature of the rocks that lie at depth below the sedimentary succession: in general terms, oceanic crust retains a higher remnant magnetism than continental crust, allowing the crustal substrate of a basin to be determined. The strength of the Earth's magnetic field at any point on the surface depends on the density of the rocks below the surface at that point. Gravity surveys can therefore provide an indication of the thickness of sedimentary strata present, as they are of lower density than igneous or metamorphic rocks. Geophysical surveys are therefore a useful way of distinguishing between different basin types (e.g. extensional backarc basins are floored by oceanic crust) and the amount of subsidence that has occurred (basins in strike-slip setting commonly have very thick sedimentary successions).

Thermal history

Burial of sediments results in diagenetic changes that include the effects of increased temperature with burial depth. The temperature that a body of sediment has been subjected to can be determined by fission-track analysis and by studying the vitrinite reflectance characteristics of organic matter present in the sediment. The burial history of a body of sediment can therefore be reconstructed using these palaeothermometers, as they record the maximum temperature that the material has reached, and this can be used to infer the depth to which it has been buried. Combining these burial history data with the age of the strata can provide information on the history of subsidence in the basin and this can be related to the basin setting.

Stratigraphic analysis

These provide a time framework for the basin history, indicating when the basin first started to form (the age of the rocks that lie at the bottom of the basin), and when sedimentation ceased (the youngest strata preserved), as well as events in between. The rate of sediment accumulation, that is, the thickness of strata deposited between two datable horizons, can be a characteristic indicator of basin setting: for example, rift basin sediments will commonly accumulate at a faster rate than passive margin deposits. On a shorter time scale, changes in sediment accumulation rate may reflect the relative sea level.

Sedimentological analysis

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The nature and the distribution of sediment present in a succession will reflect the basin setting. Three main aspects of the sedimentological analysis of basins can be considered: provenance studies, the distribution of facies and palaeoenvironments, and the changes in these through time during the basin evolution.

Provenance studies are a key element of the analysis of a basin, providing information about the tectonic setting. Rift basins in continental crust contain material derived from the surrounding cratonic area and are likely to include clasts of plutonic igneous or metamorphic origin. Peripheral foreland basins normally contain a high proportion of reworked sedimentary rocks that have been uplifted and subsequently eroded as part of the mountain-building process.

The distribution of facies within an individual package provides a picture of the distribution of palaeoenvironments for that time interval, and hence a palaeogeography can be established. Different basin types can be expected to show different patterns of sedimentation: for example, a rift or strike-slip basin may be expected to have coarse facies such as alluvial fans or fan-deltas at its margins, a backarc basin would have an apron of volcaniclastic deposits at one margin, and a passive margin succession would be dominated by shallow marine clastic or carbonate facies.

The tectonic setting of the basin is a major factor controlling the changes in the facies distributions and palaeogeographical patterns through time. Changing palaeogeography within a basin therefore reflects the tectonic evolution of both the basin and the surrounding area.

Geohistory analysis

The quantitative study of the history of subsidence and sedimentation in a basin is known as geohistory analysis (Allen & Allen 2005). As sediment accumulates, the material in the lower part of the succession undergoes compaction, so the thickness of each stratal unit decreases through time. The compaction effect varies considerably with different lithologies, with clay-rich sediments decreasing in volume by up to 80% through time, whereas sandstones typically lose between 10 and 20% of their porosity as a result of compaction. The history of subsidence in a basin can be calculated by 'decompacting' the sedimentary succession in a series of stages and by taking into account the palaeobathymetry, the water depth at which the sedimentation occurred determined from facies and palaeontological studies.

Geohistory analysis is important in hydrocarbon exploration because it provides information on the porosity and permeability changes through time, and also the thermal history of any part of the succession, which is critical to the generation of oil and gas.

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